Widget vocabulary, against real contracts.

V1 sketched shells with a fake block model. V2 is grounded: every shell in this doc consumes a single fixture shaped exactly like `RuntimeEnvelope.fragments` and `ProjectedState.sections`. The matrix below is the rendering contract; the triptych proves it round-trips across three very different UIs; the spec cards fix required fields, states, keyboard, and fallbacks per widget; the parity table extends the vocabulary to CLI, Ren'Py/Godot, and a terminal client.

■ required field ☐ optional field ■ hint / staging source: tangl.journal.fragments + tangl.service.response fixture: crossroads_inn · turn 03

§1 FRAGMENT × WIDGET × SHELL MATRIX

one row per canonical fragment_type / value_type

FRAGMENT / VALUE_TYPE	WIDGET	SCROLL	DOSSIER	STAGE + LOG
RUNTIMEENVELOPE.FRAGMENTS				
content	Prose block	inline paragraph	stage prose column	caption (if current) + log row
attributed	Dialog line	indented line w/ avatar	bracketed under prose	active caption; log row
media · cover_im	Banner frame	top banner	persistent header	fullscreen bg
media · narrative_im	Inline figure	inline w/ wrap	stage column	set dressing
media · avatar_im	Avatar chip	beside dialog line	beside dialog line	character sprite
media · dialog_im	Inline figure	attached to line	attached to line	lower-left inset
media · audio/video	Player	inline controls	footer controls	overlay HUD
choice	Button / input	bottom list	bottom of stage column	tray below log
group · scene	(implicit turn)	(boundary)	(boundary)	scene change
group · dialog	Indented rule	border-left rule	bracketed block	consecutive captions
group · turn	(implicit)	separator	separator	log divider
group · overlay	Modal sheet	sticky sheet	right-rail takeover	dim + center
group · status_sidecar	KV rail	collapsible strip	right rail	HUD top-left
kv	KV strip	inline strip	merge into rail	HUD
control · update	(silent)	replace target	replace target	replace target
control · delete	(silent)	remove target	remove target	remove target
user_event	Toast / stash	toast	toast	HUD blip
PROJECTEDSTATE.SECTIONS (SIDECAR)				
scalar	Big number / badge	strip cell	rail tile	HUD value
kv_list	Key-value table	strip (wrapped)	rail rows	HUD stack
item_list	Item roster	drawer	rail list	inventory overlay
table	Data table	drawer	rail mini-table	overlay sheet
badges	Tag strip	inline chips	rail chips	HUD chips

Rule of thumb. The shell is allowed to choose *where* a widget lives, not *whether* it renders. If a client can't render a fragment type, it must fall back to a textual rep listed in the parity table — never silently drop, because `control` fragments target UIDs and later turns assume those UIDs still resolve.

§2 SAME PAYLOAD, THREE SHELLS

crossroads_inn · turn 03

All three panels below read the *same* `window.FIXTURE`. The only thing that differs is how each shell maps `group`, `media.staging_hints.media_position`, and the projected-state sections onto screen geometry.

SCROLL — journal-style, linear

COVER_IM — BANNER · TOP · FADE_IN

Rain drums on the thatch. Inside the Crossroads Inn the air is smoke and peat. A man in a rust-colored cloak beckons from the corner booth, a folded vellum square weighted under his cup.

NARRATIVE_IM — PORTRAIT · RIGHT

AVA **Stranger** *(low, confidential)*
They say you're headed north. I've got something the garrison would pay for and never use.

? **Stranger** *(smiling without teeth)*
Forty silver. You look at it, you buy it. No haggle.
[dialog_im · generating…]

time: **late evening** coin: **63 s** companions: **Bram, Elen** weather: **rain**

1 Pay the forty silver.

2 Tell him to keep his paper.

3 Haggle. — ACCEPTS · INTEGER

4 ~~Lift the map while he drinks.~~ — Requires Sleight of Hand ≥ 2

COVER_IM — CROSSROADS_INN_NIGHT.JPG

Rain drums on the thatch. Inside the Crossroads Inn the air is smoke and peat. A man in a rust-colored cloak beckons from the corner booth, a folded vellum square weighted under his cup.

> **Stranger** *(low, confidential)* They say you're headed north. I've got something the garrison would pay for and never use.
> **Stranger** *(smiling without teeth)* Forty silver. You look at it, you buy it. No haggle.

1. Pay the forty silver.
2. Tell him to keep his paper.
3. Haggle. — integer
4. Lift the map while he drinks. — Requires Sleight of Hand ≥ 2

WOUNDS — SCALAR

Sound

PURSE — KV_LIST

silver	63
copper	11
favors	1

SATCHEL — ITEM_LIST

Hooded lantern
half-oil
light

Letter, sealed
for Captain Ros
quest

Elen's knife
balanced
weapon borrowed

PARTY — TABLE

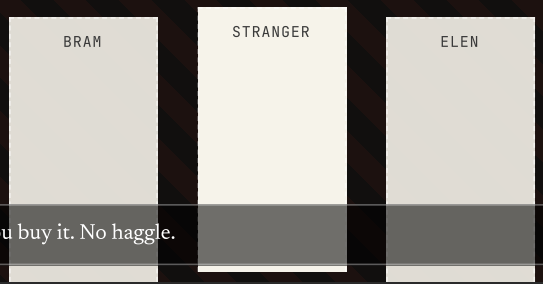
name	role	mood
Bram	soldier	surly
Elen	scout	watchful

CONDITIONS — BADGES

RAIN-SOAKED HUNGRY HUNTED

STAGE + LOG VN-ish + fragment tape

SCENE · CROSSROADS_INN · RAIN · NIGHT



STRANGER — SMILING WITHOUT TEETH Forty silver. You look at it, you buy it. No haggle.

+0:00	MEDIA	cover_im url
+0:01	CONTENT	Rain drums on the thatch. Inside the Crossroads Inn the air …
+0:02	MEDIA	narrative_im url
+0:03	GROUP	dialog × 4
+0:04	KV	4 keys
+0:05	CHOICE	Pay the forty silver.
+0:06	CHOICE	Tell him to keep his paper.
+0:07	CHOICE	Haggle.
+0:08	CHOICE	Lift the map while he drinks.

1 Pay the forty silver.

2 Tell him to keep his paper.

3 Haggle.

4 Lift the map while he drinks.

What each shell decides. *Scroll* treats the scene group as a chapter and the kv fragment as inline chrome; *Dossier* routes all projected sections to the right rail and keeps the kv inline with the prose; *Stage+Log* elevates the most recent attributed fragment to a caption and renders every fragment as a row on the tape — useful for replay and debugging.

PROSE BLOCK

CONTENT

Rain drums on the thatch. Inside the inn the air is smoke and peat…

REQ	`content`
OPT	`format` `hints.style_tags` `hints.style_dict` `source_id`
FALLBACK	If format unknown, render content as plain text; ignore hints.
MOBILE	full width, serif body, 16px min
KEYBOARD	Tab skips; selectable text

EMPTY skip

LOADING stream text in as chunks arrive

ERROR render raw string with a red margin-mark

DIALOG LINE

ATTRIBUTED

S **Stranger** *(low)*
Forty silver. No haggle.

REQ	`who` `how` `media` `content`
OPT	`hints`
FALLBACK	Render as prose prefixed with who:
MOBILE	avatar 32px, who above body
KEYBOARD	skippable via arrow keys in replay

EMPTY hide line entirely

LOADING placeholder avatar, ellipsis body

ERROR show who + raw content; drop how

MEDIA FRAME

MEDIA

narrative_im

REQ	content content_format
OPT	media_role staging_hints.* scope
FALLBACK	If role unknown → inline image; if content_format=rit and unresolved → placeholder card with RIT id.
MOBILE	full bleed for cover_im; inline for narrative_im
KEYBOARD	focusable for longform media (audio, video); space to toggle

EMPTY hide

LOADING striped placeholder + role label

ERROR striped placeholder + error text; keep layout

CHOICE BUTTON / INPUT

CHOICE

1 Pay the forty silver.

4 ~~Lift the map.~~ SOH ≥ 2

REQ	edge_id text
OPT	available unavailable_reason blockers[] accepts{} ui_hints{hotkey,icon,emphasis} payload
FALLBACK	Render disabled if unknown; log a warning; never silently drop.
MOBILE	44px hit target; stacked full-width
KEYBOARD	1-9 hotkey from ui_hints; ↑↓ cycles; Enter commits; Esc cancels freeform

EMPTY must have ≥1 available for turn to be actionable — emit blocker notice

LOADING disable entire choice group; show spinner near primary

ERROR mark action as failed; keep choices enabled for retry

CONTAINER (ROLE-SWITCHED)

GROUP

group_type ∈ {scene, dialog, turn, overlay, status_sidecar}

REQ	group_type member_ids[]
OPT	hints
FALLBACK	If group_type unknown → render members in order with no wrapper.
MOBILE	dialog group = indented rule; scene group = implicit; overlay = sheet
KEYBOARD	dialog group is aria-live region; overlay traps focus

EMPTY hide

LOADING render members in order as they arrive

ERROR render members flat; warn

KV STRIP OR SIDE RAIL

KV

time: **late** coin: **63** weather: **rain**

REQ	content (OrderedTupleDict)
OPT	hints.style_tags (e.g. status-inline)
FALLBACK	Render as 'k: v' plaintext lines.
MOBILE	wrap into 2-col; rail collapses into top strip
KEYBOARD	not focusable by default (informational)

EMPTY hide

LOADING skeleton rows

ERROR render known pairs; mark failed ones

(SILENT) FRAGMENT MUTATION

CONTROL

update → content#glossary:vellum

REQ	ref_type ref_id fragment_type ∈ update|delete
OPT	payload (required for update)
FALLBACK	Ignore control if ref_id not found in local registry; log.
MOBILE	invisible — triggers re-render of target
KEYBOARD	invisible

EMPTY n/a

LOADING n/a — applied atomically

ERROR retain original fragment; surface to author view only

TOAST / SILENT STASH

USER_EVENT

★ met_10_strangers · 7/10

REQ	event_type
OPT	content (any)
FALLBACK	Unknown event_type → stash on user record, no UI.
MOBILE	bottom toast, auto-dismiss 3s
KEYBOARD	Esc dismisses

EMPTY skip

LOADING n/a

ERROR silent; log to author view

§4 EDGE CASES WORTH FIXING NOW

before per-world clients fork

LOCKED CHOICES

`available=false` + `unavailable_reason` is the user-visible contract.`blockers[]` is structured — author tools show it, players don't. Render as a disabled button with reason as secondary text; keep it in the DOM so screen readers and keyboard order are stable.

FREEFORM INPUT CHOICES

`accepts` turns a choice into a form. Widget = labeled input inline with the button. Enter commits `payload`; Esc reverts to a plain button. CLI renders a `'>'` prompt after the numbered list.

PENDING GENERATED MEDIA

When `content_format=rit` and unresolved, emit the media fragment anyway with a stable UID. Client renders the striped placeholder with role + RIT id. A later `control·update`swaps in a `url` or `data` payload — same widget, same DOM node, no reflow.

MULTIPLE MEDIA ROLES IN ONE TURN

Shells route by `media_role`, not order. cover_im is persistent chrome; narrative_im belongs to the active content; avatar_im + dialog_im bind to the nearest preceding attributed fragment; audio/video/sfx are timelined against `staging_hints.media_timing`.

GROUP SEMANTICS

Treat `group_type` as the rendering hint, not `fragment_type`. Required set: `scene, dialog, turn, overlay, status_sidecar`. Unknown group_type → render members flat. Nesting is allowed one level (overlay inside scene).

KV: FLOW VS RAIL VS BOTH

A `KvFragment` in the stream is scene-bound (e.g. "time: late, weather: rain"). A projected `kv_list` section is durable (purse, stats). Scroll renders both inline; Dossier promotes projected to the rail and keeps stream-kv inline; CLI flattens everything to a status line.

"CURRENT", "HISTORY", "PERSISTENT"

Current = fragments with `cursor_id` == envelope.cursor_id. History = prior envelopes, dimmed, choices disabled. Persistent = projected_state, re-projected every turn; shell is free to animate deltas.

A11Y & REDUCED MOTION

Choice group is a `role=group` with `aria-label`; hotkeys come from `ui_hints.hotkey`; focus returns to the primary choice after an action resolves.`prefers-reduced-motion` disables `media_transition` and swaps to instant. Stage captions obey an `aria-live=polite` region.

§5 AUTHOR / DEBUG MODE

grounded in envelope + ledger + phase bus

PHASES

BUS

```
resolve → compose → project → dispatch
step 47 · cursor b6e0f3a4…
last_redirect: null
redirect_trace: []
```

EMITTED FRAGMENTS

11

```
1 group·scene
1 content
2 media (cover, narr)
1 group·dialog
2 attributed
2 media (avatar, dlg·pending)
1 kv · 4 choices · 1 user_event · 1 control·update
```

PROJECTED SECTIONS

5

```
scalar · Wounds
kv_list · Purse
item_list · Satchel
table · Party
badges · Conditions
```

LEDGER / GRAPH

REF

```
current node: crossroads_inn · ch01·03
traversed edges: 46
open edges: 4 (1 locked, 1 accepts payload)
blockers: sleight_of_hand≥2 (have 1), elen_watching
```

§6 PORT PARITY

same vocabulary in web / CLI / Ren'Py / terminal

WIDGET	WEB (VUE / BESPOKE)	CLI (CURRENT)	REN'PY / GODOT	TCL / TERMINAL
Prose block	<article>	wrapped lines	narrator say	reflowed stdout
Dialog line	avatar + bubble	who: text	character say	'who> text'
Cover/narr media	<img>/<video>	[img: url]	scene bg / show expr	link + ascii box
Avatar	round <img>	(elided)	side image	(elided)
Audio	<audio>	[♪ url]	play music / sound	[♪]
Choice	button list	1) … 2) …	menu:	numbered prompt
Locked choice	disabled + reason	(locked) reason	if-gated, grey	greyed
Freeform choice	inline input	prompt '>'	input / pyexpr	readline
Dialog group	indented rule region	blank lines bracket	contiguous say block	blank-line bracket
Overlay group	modal sheet	'---' banner	screen call with modal	full-screen page
KV rail	right rail	[status] line	stats screen	status bar
Scalar / badge	tile	key=value	text widget	status seg
Item list	roster	'- item (detail)'	inventory screen	list cmd
Table	<table>	aligned cols	grid()	aligned cols
Toast / user_event	bottom toast	'* note' line	notify()	status flash
control update/delete	re-render target	re-print with marker	re-show statement	re-print

Bespoke per-world clients. The vocabulary contract means a custom client can *add* widgets (card-table, dice tray, map) without breaking the core stream — it opts in to render specific `group_types` or `ui_hints.widget` values, and falls through to the parity row for everything else.

— end · v2 wireframes · sources: tangl.journal.fragments (ContentFragment, MediaFragment, ChoiceFragment, GroupFragment, KvFragment, ControlFragment, UserEventFragment, AttributedFragment, DialogFragment), tangl.service.response (RuntimeEnvelope, ProjectedState, ScalarValue, KvListValue, ItemListValue, TableValue, BadgeListValue, PresentationHints, StagingHints)